ST5202 — Applied Regression Analysis
Lecture Notes 0. Preliminary materials: a review

Chen Zehua

Department of Statistics and Data Science

Outline

§1. Results on linear algebra

§2. Review on basic statistical topics

§3. A brief introduction to R

§1.1. Scalar, Vector and Matrix

- A quantity is called a Scalar, if the quantity has only a single value.
- A finite sequence of scalars arranged either in a row or a column is called a Vector (row or column vector). The number of components in a vector is called its dimension. A vector of dimension n is referred to as a n-vector. A n-vector $\boldsymbol{x}$ is denoted by

$$\boldsymbol{x} = (x_1, x_2, \cdots, x_n)^\top.$$

- Scalars arranged in a formation of m rows and n columns is called a $m \times n$ matrix where m and n are, respectively, its row and column dimension. A $m \times n$ matrix A is denoted by

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & \cdots & \cdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

- A scalar is a 1×1 matrix, a special case of matrix.
- A vector is a special matrix of only one row or one column.

§1.2. Matrix calculation

- A $m \times n$ matrix A is abbreviated as
$$A = (a_{ij})_{i=1,\ldots,m;j=1,\ldots,n} = (a_{ij}),$$
where a_{ij} denotes the entry of A in the ith row and jth column.
- The transpose of $A = (a_{ij})$ is given by $A^\top = (a_{ji})$.
- If A and B have the same dimensions,
$$A \pm B = (a_{ij} \pm b_{ij}).$$
- If c is a scalar, $$cA = (ca_{ij}).$$
- Let $\boldsymbol{a}$ and $\boldsymbol{b}$ be two n-vectors. The inner product of $\boldsymbol{a}$ and $\boldsymbol{b}$ is defined as
$$\boldsymbol{a}^\top \boldsymbol{b} = \sum_{i=1}^{n} a_i b_i.$$
If $\boldsymbol{a}^\top \boldsymbol{b} = 0$, $\boldsymbol{a}$ and $\boldsymbol{b}$ are said to be perpendicular to each other.
- Let $A = (a_{ij})$ be a $m \times n$ matrix and $B = (b_{jk})$ be a $n \times p$ matrix. Their matrix product AB is said to be compatible and is defined as
$$AB = \left(c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk} \right).$$

§1.3. Rank and trace

- Let $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p$ be p n-vectors. A linear combination of them is given by
$$\sum_{j=1}^{p} c_j \boldsymbol{x}_j = c_1 \boldsymbol{x}_1 + \cdots + c_p \boldsymbol{x}_p,$$
where c_j are scalars.
- If $\sum_{j=1}^{p} c_j \boldsymbol{x}_j = 0$ implies $c_j = 0$ for all j, the vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p$ are said to be linearly independent, otherwise, linearly dependent. In the latter case, any one with non-zero coefficient can be expressed as a linear combination of the others.
- The maximum number of independent columns (or equivalently, indpendent rows) of a matrix A is called its rank, and is denoted by $rk(A)$.
- If $A = (a_{ij})$ is a square matrix ($m = n$), the sum of its diagonal elements, $\sum a_{ii}$, is called its trace, and denoted by $tr(A)$.

§1.4. Determinant, eigenvalue and eigenvector

- The determinant $|A|$ of a square matrix $A = (a_{ij})$ is defined inductively as

$$|A| = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} |A_{ij}|,$$

  where A_{ij} is the matrix obtained from A by deleting its ith row and jth column.

- Let A be an $n \times n$ square matrix. The determinant $|A - \lambda I|$ is a polynomial of λ with order n. The roots of this polynomial are called the eigenvalues of A.

- Let λ be an eigenvalue of A, any vector $\boldsymbol{y}$ satisfying

$$A\boldsymbol{y} = \lambda \boldsymbol{y}$$

  is called an eigenvector of A corresponding to eigenvalue λ. If $\boldsymbol{y}$ is such that $\sum_{i=1}^{n} y_i^2 = 1$, it is called the standardized eigenvector.

§1.5. Special matrices

- A square matrix whose diagonal entries are 1 and off-diagonal entries are zero is called a identity matrix and denoted by I.
- A $n \times n$ matrix B is called the inverse matrix of a $n \times n$ matrix A, if $AB = BA = I$. B is denoted as A^{-1}.
- A matrix A is symmetric if $A = A^\top$.
- A square matrix A is idempotent if $A = A^2$.
- If A is symmetric and, for any vector $\boldsymbol{x}$, $\boldsymbol{x}^\top A\boldsymbol{x} > (<)0$, A is said to be positive (or negative) definite. When $>$ or $<$ are replaced by $\geq$ or $\leq$, A is said to be non-negative (or non-positive) definite.
- If A is a square matrix and $A^\top A = I$, A is called an orthogonal matrix.

§1.6. Some useful results

- If A is idempotent, then
  (a) the eigenvalues of A are 1 or 0.
  (b) $rk(A) = tr(A)$.
- The eigenvalues of a positive (or negative) definite matrix are always positive (or negative).
- If AB and BA are both compatible, then
  (a) $tr(AB) = tr(BA)$.
  (b) AB and BA have the same non-zero eigenvalues.
- If A is non-negative definite,

$$\max_{\boldsymbol{x}} \frac{\boldsymbol{x}^\top A \boldsymbol{x}}{\boldsymbol{x}^\top \boldsymbol{x}} = \lambda_{\max}(A),$$

  where $\lambda_{\max}$ denotes the largest eigenvalue.
- If A is non-negative definite, B is positive-definite,

$$\max_{\boldsymbol{x}} \frac{\boldsymbol{x}^\top A \boldsymbol{x}}{\boldsymbol{x}^\top B \boldsymbol{x}} = \lambda_{\max}(AB^{-1}).$$

§1.7. Matrix decomposition (1)

- **Cholesky decomposition**. If A is non-negative definite, it can be decomposed as

$$A = R^\top R.$$

  If R is an upper triangular matrix with non-negative diagonal elements, such decomposition is unique and is called the Cholesky decomposition.

- **QR decomposition**. For any $n \times p$ $(n > p)$ matrix A, there is an orthogonal matrix Q such that

$$A = QR,$$

  where R is a $n \times p$ matrix with all elements zero below its diagonal, i.e., $r_{ij} = 0$, for $i > j$.

§1.8. Matrix decomposition (2)

- **Singular value decomposition**. For any $n \times p$ matrix A, there is an $n \times n$ orthogonal matrix U and a $p \times p$ orthogonal matrix V such that

$$U^\top AV = \begin{pmatrix} \Sigma \\ 0 \end{pmatrix}, \qquad \text{if } n \geq p,$$

$$U^\top AV = \begin{pmatrix} \Sigma & 0 \end{pmatrix}, \qquad \text{if } n \leq p.$$

  Here $\Sigma = diag(\sigma_1, \sigma_2, \ldots, \sigma_m)$, where $m = \min\{n, p\}$ and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_m > 0$.

- **Eigenvalue-eigenvector decomposition**. For any symmetric matrix A, there is an orthogonal matrix Q such that

$$A = Q^\top DQ,$$

  where D is a diagonal matrix whose diagonal elements are the eigenvalues of A. The columns of Q are the corresponding standardized eigenvectors.

§1.9. Vector space (1)

- The collection of all n-vectors is called the Euclidean space, denoted by R^n.
- For $\boldsymbol{x}_j \in R^n, j = 1, \ldots p$ $(p < n)$, the collection of all possible linear combinations $\sum_{j=1}^{p} c_j \boldsymbol{x}_j$ form a subspace of R^n. This subspace is called the space spanned by $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p$.
- If $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p$ are not linearly independent and the largest number of linearly independent vectors among them is d, then any subset of $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p$ containing d linearly independent vectors span the same subspace. The number d is referred to as the dimension of the subspace.

§1.10. Vector space (2)

- The distance between any two vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ is defined as $\|\boldsymbol{x}-\boldsymbol{y}\|_2 = \sqrt{(\boldsymbol{x}-\boldsymbol{y})^\top(\boldsymbol{x}-\boldsymbol{y})}$. The projection of a vector $\boldsymbol{y} \in R^n$ into a subspace is the vector in that subspace which has the smallest distance from $\boldsymbol{y}$.
- Suppose a subspace is spanned by linearly independent vectors $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_p$ and X is the matrix consisting of these vectors as its columns. The matrix $H = X(X^\top X)^{-1}X^\top$ is called the projection matrix of the subspace. For any vector $\boldsymbol{y}$, its projection into the subspace is given by $H\boldsymbol{y}$.
- For any d-dimensional subspace of R^n, say R_1, there is a $(n-d)$-dimensional subspace of R^n, say R_2, such that for any $\boldsymbol{x} \in R_1$ and $\boldsymbol{y} \in R_2$, $\boldsymbol{x}^\top\boldsymbol{y} = \sum_{i=1}^n x_i y_i = 0$, i.e., $\boldsymbol{x}$ and $\boldsymbol{y}$ are perpendicular to each other.

§1.11. Matrix calculus (1)

- *Matrix form of linear functions:*

$$\sum_{i=1}^{n} a_i x_i = a_1 x_1 + \cdots + a_n x_n = \boldsymbol{a}^\top \boldsymbol{x},$$

where $\boldsymbol{a} = (a_1, \ldots, a_n)^\top$, $\boldsymbol{x} = (x_1, \ldots, x_n)^\top$.

- *Matrix form of quadratic functions:*

$$\sum_{i=1}^{n} \sum_{j=1}^{n} a_{ij} x_i x_j = \boldsymbol{x}^\top A \boldsymbol{x},$$

where

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \cdots & \cdots & \cdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}.$$

§1.12. Matrix calculus (2)

Let $f(\boldsymbol{x})$ be a multivariate function. Define

$$\frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{x}} = \left(\frac{\partial f(\boldsymbol{x})}{\partial x_1}, \ldots, \frac{\partial f(\boldsymbol{x})}{\partial x_n} \right)^\top.$$

(i) $\frac{\partial \boldsymbol{a}^\top \boldsymbol{x}}{\partial \boldsymbol{x}} = \boldsymbol{a}$. E.g. for $f(\boldsymbol{x}) = 2x_1 + 3x_2$,

$$\frac{\partial f(\boldsymbol{x})}{\partial x_1} = 2, \quad \frac{\partial f(\boldsymbol{x})}{\partial x_2} = 3,$$

$$\frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{x}} = \left(\frac{\partial f(\boldsymbol{x})}{\partial x_1}, \frac{\partial f(\boldsymbol{x})}{\partial x_2} \right)^\top = (2, 3)^\top = \boldsymbol{a}.$$

§1.13. Matrix calculus (3)

(ii) $\frac{\partial \boldsymbol{x}^\top A\boldsymbol{x}}{\partial \boldsymbol{x}} = 2A\boldsymbol{x}$ if A is symmetric, $(A + A^\top)\boldsymbol{x}$ otherwise.
E.g. for

$$A = \begin{pmatrix} 3 & 4 \\ 4 & 5 \end{pmatrix}, \ f(\boldsymbol{x}) = \boldsymbol{x}^\top A\boldsymbol{x} = 3x_1^2 + 2 \times 4x_1x_2 + 5x_2^2,$$

we have

$$\frac{\partial f(\boldsymbol{x})}{\partial x_1} = 2 \times 3x_1 + 2 \times 4x_2 = 2(3x_1 + 4x_2)$$

$$\frac{\partial f(\boldsymbol{x})}{\partial x_2} = 2 \times 4x_1 + 2 \times 5x_2 = 2(4x_1 + 5x_2)$$

$$\frac{\partial f(\boldsymbol{x})}{\partial \boldsymbol{x}} = \begin{pmatrix} 2(3x_1 + 4x_2) \\ 2(4x_1 + 5x_2) \end{pmatrix} = 2\begin{pmatrix} 3 & 4 \\ 4 & 5 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2A\boldsymbol{x}.$$

§1.14. Inverse of blocked matrix

- Let Σ be partitioned as

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}.$$

  The partition is called a blocked matrix.

- If Σ is non-singular and both Σ_{11} and Σ_{22} are square matrices, then

$$\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}^{-1} = \begin{pmatrix} \Sigma_{11}^{-1} + \Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22.11}^{-1}\Sigma_{21}\Sigma_{11}^{-1} & -\Sigma_{11}^{-1}\Sigma_{12}\Sigma_{22.11}^{-1} \\ -\Sigma_{22.11}^{-1}\Sigma_{21}\Sigma_{11}^{-1} & \Sigma_{22.11}^{-1} \end{pmatrix}$$

  where
$$\Sigma_{22.11} = \Sigma_{22} - \Sigma_{21}\Sigma_{11}^{-1}\Sigma_{12}.$$

- Also

$$\begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}^{-1} = \begin{pmatrix} \Sigma_{11.22}^{-1} & -\Sigma_{11.22}^{-1}\Sigma_{12}\Sigma_{22}^{-1} \\ -\Sigma_{22}^{-1}\Sigma_{21}\Sigma_{11.22}^{-1} & \Sigma_{22}^{-1} + \Sigma_{22}^{-1}\Sigma_{21}\Sigma_{11.22}^{-1}\Sigma_{12}\Sigma_{22}^{-1} \end{pmatrix}$$

  where
$$\Sigma_{11.22} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}.$$

§2.1. Population and distribution

- A population is a collection of individual subjects (finite or infinite).
- If each individual in a population has a numerical characteristic X, then X has a distribution. Roughly, the distribution describes the proportions of any values or set of values of X in the population.
- The numerical characteristic X is called a random variable. The cumulative distribution function (cdf) of X is defined as $F(x) = P(X \leq x)$, where $P(X \leq x)$ is the proportion of the subjects with $X \leq x$, and it is referred to as the probability of $X \leq x$.
- If F is differentiable, the probability density function of X is defined as

$$f(x) = dF(x)/dx.$$

- If X takes only a finite number of values, say $x_1, x_2, \cdots, x_m$, the probability density of X is defined as

$$p(x_i) = P(X = x_i), i = 1, \cdots, m,$$

where $P(X = x_i)$ is the proportion of the subjects with $X = x_i$.

§2.2. Basic characteristics of random variables (1)

- The mean (or expectation) of X:

$$\mu = EX = \int x dF(x).$$

  The mean, in a broad sense, is the average of the X values of the individuals in the population.

- The variance of X:

$$\sigma^2 = \mathrm{Var}(X) = E(X - EX)^2.$$

  The variance describes how spread the values of X are.

- The moments of X:

$$m_j = EX^j = \int x^j dF(x).$$

- The quantile of X: Suppose $f(x)$ is continuous. The $(1-\alpha)$-quantile (or the upper α-quantile), q_α, is defined such that

$$F(q_\alpha) = 1 - \alpha, \text{ or } 1 - F(q_\alpha) = \alpha.$$

§2.3. Basic characteristics of random variables (2)

- Let X and Y be two random variables of the same population and $F(x, y)$ be their joint distribution function defined as

$$F(x, y) = P(X \leq x, Y \leq y).$$

The covariance between X and Y is defined as

$$\mathrm{Cov}(X, Y) = E[(X - EX)(Y - EY)] = \int (x - EX)(y - EY) dF(x, y).$$

- The Pearson's correlation coefficient between X and Y is defined as

$$\rho_{xy} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}.$$

§2.4. Random vectors

- A vector of random variables is called a random vector.
- Let $\boldsymbol{x} = (X_1, \ldots, X_p)^\top$, $\boldsymbol{y} = (Y_1, \ldots, Y_q)^\top$. The mean, variance and convariance are defined as follows.

$$
\begin{aligned}
E\boldsymbol{x} &= (EX_1, \ldots, EX_p)^\top \\
\mathrm{Var}(\boldsymbol{x}) &= \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots & \mathrm{Cov}(X_1, X_p) \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots & \mathrm{Cov}(X_2, X_p) \\ \cdots & \cdots & \cdots & \cdots \\ \mathrm{Cov}(X_p, X_1) & \mathrm{Cov}(X_p, X_2) & \cdots & \mathrm{Var}(X_p) \end{pmatrix} \\
\mathrm{Cov}(\boldsymbol{x}, \boldsymbol{y}) &= \begin{pmatrix} \mathrm{Cov}(X_1, Y_1) & \mathrm{Cov}(X_1, Y_2) & \cdots & \mathrm{Cov}(X_1, Y_q) \\ \mathrm{Cov}(X_2, Y_1) & \mathrm{Cov}(X_2, Y_2) & \cdots & \mathrm{Cov}(X_2, Y_q) \\ \cdots & \cdots & \cdots & \cdots \\ \mathrm{Cov}(X_p, Y_1) & \mathrm{Cov}(X_p, Y_2) & \cdots & \mathrm{Cov}(X_p, Y_q) \end{pmatrix}
\end{aligned}
$$

§2.5. Formulae for mean, variance and covariance

- For random variables:
  - (i) $E(\sum_{j=1}^{p} c_j X_j) = \sum_{j=1}^{p} c_j EX_j$.
  - (ii) $\mathrm{Var}(\sum_{j=1}^{p} a_j X_j) = \sum_{i=1}^{p} \sum_{j=1}^{p} a_i a_j \mathrm{Cov}(X_i, X_j)$, in general;
  - (iii) $\mathrm{Var}(\sum_{j=1}^{p} a_j X_j) = \sum_{j=1}^{p} a_j^2 \mathrm{Var}(X_j)$, if X_j's are un-correlated.
  - (iv) $\mathrm{Cov}(\sum_{i=1}^{p} a_i X_i, \sum_{j=1}^{q} b_j Y_j) = \sum_{i=1}^{p} \sum_{j=1}^{q} a_i b_j \mathrm{Cov}(X_i, Y_j)$.
- For random vectors: Let A, B and $\boldsymbol{b}$, $\boldsymbol{c}$ be constant matrices and vectors. Let $\boldsymbol{x}$ and $\boldsymbol{y}$ be random vectors.
  - (i) $E(A\boldsymbol{x} + \boldsymbol{b}) = AE\boldsymbol{x} + \boldsymbol{b}$,
  - (ii) $\mathrm{Var}(A\boldsymbol{x} + \boldsymbol{b}) = A\mathrm{Var}(\boldsymbol{x})A^\top$.
  - (iii) $\mathrm{Cov}(A\boldsymbol{x} + \boldsymbol{b}, B\boldsymbol{x} + \boldsymbol{c}) = A\mathrm{Var}(\boldsymbol{x})B^\top$.
  - (iv) $\mathrm{Cov}(A\boldsymbol{x} + \boldsymbol{b}, B\boldsymbol{y} + \boldsymbol{c}) = A\mathrm{Cov}(\boldsymbol{x}, \boldsymbol{y})B^\top$.

§2.6. Random sample and empirical distribution (1)

- The population characteristics are unknown. The role of Statistics is to make inference on the population characteristics through random samples taken from the population.
- A random sample of size n is n individuals taken from the population at random with replacement. The X values of the random sample is denoted by $(X_1, \ldots, X_n)$, their observed values are denoted by $(x_1, \ldots, x_n)$.
- The empirical distribution of X is defined as

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} I\{x_i \leq x\},$$

where $I\{x_i \leq x\} = 1$ if $x_i \leq x$, 0, otherwise.

§2.7. Random sample and empirical distribution (2)

The sample counterparts of the population characteristics are the following:

- Sample mean: $\hat{\mu} = \bar{X} = \frac{1}{n}\sum_{i=1}^{n} x_i$.
- Sample variance: $\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{X})^2$.
- Sample moments: $\hat{m}_j = \frac{1}{n}\sum_{i=1}^{n} x_i^j$.
- Sample covariance: $\frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{X})(y_i - \bar{Y})$.
- Sample Pearson's correlation coefficient:

$$\hat{\rho}_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{X})(y_i - \bar{Y})}{[\sum_{i=1}^{n}(x_i - \bar{X})^2 \sum_{i=1}^{n}(x_y - \bar{Y})^2]^{1/2}}.$$

An important statistical principle is to make inference on the population characteristics by using their sample counterparts.

§2.8. Uni-variate Normal distribution

- A random variable X has an uni-variate normal distribution if it has p.d.f. given by
$$f(y|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(y-\mu)^2},$$
where μ is the mean and σ^2 is the variance of X. The normal distribution is denoted by $N(\mu,\sigma^2)$. If $\mu = 0, \sigma^2 = 1$, it is called the standard normal distribution.
- The standard normal distribution is symmetric about zero, i.e., for any x,
$$F(-x) = 1 - F(x),$$
or, for any $c > 0$,
$$P(|X| \geq c) = 2P(X \geq c) = 2P(X \leq -c).$$
- If $X \sim N(\mu,\sigma^2)$ then $Z = (X-\mu)/\sigma \sim N(0,1)$, where $\sim$ means "has the distribution".

§2.9. Multivariate normal distribution

- Let $\boldsymbol{y} = (Y_1, \ldots, Y_m)^\top$ be a random vector whose components are i.i.d. standard normal variables. Let A be a $q \times m$ constant matrix and $\boldsymbol{\mu}$ a constant vector. The distribution of $\boldsymbol{x} = A\boldsymbol{y} + \boldsymbol{\mu}$ is called a q-dimensional multivariate normal distribution with mean $E\boldsymbol{x} = \boldsymbol{\mu}$ and variance matrix $\Sigma = AA^\top$, and is denoted by $N(\boldsymbol{\mu}, \Sigma)$.
- The normal distribution is uniquely determined by its mean $\boldsymbol{\mu}$ and variance matrix Σ.
- If $rk(A) = q$, the pdf of $N(\boldsymbol{\mu}, \Sigma)$ is given by

$$f(\boldsymbol{x}|\boldsymbol{\mu}, \Sigma) = \frac{1}{(2\pi|\Sigma|)^{q/2}} e^{-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^\top \Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}.$$

§2.10. Properties of multivariate normal distribution

- If $\boldsymbol{x} \sim N(\boldsymbol{\mu}, \Sigma)$, then, for any constant matrix B, $B\boldsymbol{x}$ has a multivariate normal distribution $N(B\boldsymbol{\mu}, B\Sigma B^\top)$.
- In particular, for any constant vector $\boldsymbol{c}$, the linear combination $\boldsymbol{c}^\top \boldsymbol{x}$ has a uni-variate normal distribution $N(\boldsymbol{c}^\top \boldsymbol{\mu}, \boldsymbol{c}^\top \Sigma \boldsymbol{c})$.
- Any component of $\boldsymbol{x}$ is an univariate normal variable.
- If $\boldsymbol{x} \sim N(\boldsymbol{\mu}, \Sigma)$ then $\boldsymbol{z} = \Sigma^{-1/2}(\boldsymbol{x} - \boldsymbol{\mu}) \sim N(\mathbf{0}, I)$, i.e., the components of $\boldsymbol{z}$ are i.i.d. standard normal variables.
- For jointly normal-distributed random variables X and Y, that X and Y are independent is equivalent to that $\mathrm{Cov}(X, Y) = 0$.

§2.11. The χ^2- t- and F-distributions

- Let $\boldsymbol{x} = (X_1, \ldots, X_m)^\top$. Suppose X_j's are i.i.d. $N(0,1)$ variables. The distribution of $\boldsymbol{x}^\top \boldsymbol{x} = \sum_{j=1}^m X_j^2$ is called the χ^2-distribution with d.f. m and is denoted by χ^2_m.
- Suppose $Z \sim N(0,1)$, $U \sim \chi^2_m$, Z and U are independent. The distribution of $Z/(\sqrt{U/m})$ is called the t-distribution with d.f. m and is denoted by t_m.
- Suppose $U \sim \chi^2_m$, $V \sim \chi^2_n$, U and V are independent. The distribution of $\frac{U/m}{V/n}$ is called the F-distribution with d.f. m and n, denoted by $F_{m,n}$.
- If $\boldsymbol{x} \sim N(\boldsymbol{\mu}, \Sigma)$, then $W = (\boldsymbol{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\boldsymbol{x} - \boldsymbol{\mu}) \sim \chi^2_m$.
- If $\boldsymbol{x} \sim N(0, I)$, A is symmetric and idempotent, then $\boldsymbol{x}^\top A \boldsymbol{x} \sim \chi^2_r$, where $r = rk(A) = tr(A)$.

§2.12. Maximum likelihood estimation (MLE)

- If X has pdf $f(x,\theta)$, given the observation $x_1,\ldots,x_n$ of a random sample, the log likelihood function of θ is defined as
$$\ell(\theta) = \ln\left(\prod_{i=1}^{n} f(x_i,\theta)\right) = \sum_{i=1}^{n} \ln f(x_i,\theta).$$
The maximum likelihood estimate of θ is the value of θ that maximizes the log-likelihood function.
- For example, let $(x_1,\cdots,x_n)$ be the observation of a random sample from $N(\mu,\sigma^2)$. The log likelihood function of (μ,σ^2) is then
$$\begin{aligned}\ell(\mu,\sigma^2) &= \sum_{i=1}^{n} \log \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2} \\ &= -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\mu)^2.\end{aligned}$$
The MLE of μ and σ^2 are obtained by maximizing $\ell(\mu,\sigma^2)$ and are given by
$$\hat{\mu} = \bar{X},\ \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{X})^2.$$

§2.13. Confidence intervals

- A 100(1-α)% confidence interval for a parameter θ is an random interval $[L(\boldsymbol{x}),\ U(\boldsymbol{x})]$ such that

$$P([L(\boldsymbol{x}) \leq \theta \leq\ U(\boldsymbol{x})]) \geq 1 - \alpha.$$

- The most common structure of a confidence interval is

$$[\hat{\theta} - s.d(\hat{\theta})c_{\alpha/2},\ \ \hat{\theta} + s.d(\hat{\theta})c_{\alpha/2}],$$

where $\hat{\theta}$ is an estimate of θ, $s.d(\hat{\theta})$ is the estimated standard deviation of $\hat{\theta}$ and $c_{\alpha/2}$ is the upper $\alpha/2$ quantile of the distribution of $(\hat{\theta} - \theta)/s.d(\hat{\theta})$.

§2.14. Neyman-Pearson framework of hypothesis testing

- A null hypothesis H_0 and an alternative hypothesis H_1 are formulated. The two hypotheses are mutually exclusive.
- A test statistic, $T(\boldsymbol{x})$, is used to make the decision. If $T(\boldsymbol{x}) \geq c$ for a predetermined constant c, we reject H_0, otherwise, we do not reject H_0.
- In making decision this way, we might commit one of the two errors: type I error — reject H_0 when H_0 is true; type II error — do not reject H_0 when H_0 is false.
- The rate of the two errors cannot be controlled at the same time. The Neyman-Pearson framework chooses to control the type I error rate at a given level α, that is, choose c such that

$$P(T(\boldsymbol{x}) \geq c | H_0) \leq \alpha.$$

§2.15. Determination of null hypothesis

- The null H_0 and the alternative H_1 are not at the same footing, since only the type I error rate is controlled, type I error is considered more serious than type II error.
- The null hypothesis must be formulated such that the type I error is more serious.
- If it is not clear-cut which error is more serious, the general guideline is that the null hypothesis should be formulated as
  - a well-established theory;
  - the status quo;
  - the opposite of a new belief, a new finding, a guess, or a new experience.

  The guideline reflects the conservative attitude of human being towards new discoveries: unless a new discovery is sufficiently verified or there are strong evidence to support it, it should not be readily accepted.

§3.1. Installation of R and R packages

- Download and install R on your computer.
- For certain statistical procedures, specific R packages, which are not included in the basic R installation, are required. Those packages must be installed by yourself.
- You can search available R packages on the internet, and download the package to your PC and install locally.
- For example, to download and install the MASS package, follow the following steps:
  - Search `r package MASS` on Google, and find CRAN - Package MASS
  - Go to the page and download the source file.
  - Within R, follow the path `packages -- install package(s) from local file`, then select your downloaded source file.

§3.2. Basic operations

- $+$, $-$, $*$, $/$, `^`:
  Carry out component-wise addition, subtraction, multiplication, division and exponential operations. The operation applies to scalars, vectors and matrices. E.g.,
  $x = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $y = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, `x^y` $= \begin{pmatrix} 1 & 2^2 \\ 3^3 & 4^4 \end{pmatrix}$.
  If one of the operands is a scalar, the scalar is repeatedly used in the operation.
- Matrix multiplication: `x%*%y`.
- Matrix inversion: `solve(x)`.
- Transposition: `t(x)`.

§3.3. Data objects (1)

- R operates on objects. The major objects are: scalar, vector, matrix, data frame, list, etc. Values can be assigned to data objects in various ways. Components of an object can be extracted in various ways.
- Vector
  - A vector can be created by
    direct assignment: `x = c(1,2,3,4)`
    from existing vectors: `x=c(y,z)`
    from existing matrix: `x=A[,1]`, `x=A[1,]`
  - Components of a vector can be extracted
    individually: `x[2]`
    collectively: `x[1:3]`, `x[c(2,5,6)]`, `x[-c(2,4)]`

§3.4. Data objects (2)

- Matrix
  - A matrix can be created by
    direct assignment: `A=matrix(c(1,2,3,4),ncol=2,byrow=T)`
    combination of vectors: `A=cbind(x,y), A=rbind(x,y)`
  - Components of a matrix can be extracted
    individually: `A[2,3]`
    in vector: `A[1,], A[,1]`
    in matrix: `A[c(1,2),c(3,4)], A[-c(1,2), -c(3,4)]`
- Data frame. The columns of a matrix must have the same type of data, either `numeric` or `character`. A data frame is similar to a matrix, but its columns can have different types. In addition, each column is assigned a name.

§3.5. Data objects (3)

- A data frame can be created by
  - direct assignment:

    ```
    A.dat=data.frame(x=c(1,2),y=c(2,3)),
    B.dat=data.frame(age=x1, income=x2)
    ```

  - from a `.txt` file:

    ```
    A.dat=read.table("C://Rcodes/insur.txt",header=TRUE)
    ```

  - from a excel `.csv` file:

    ```
    A.dat=read.csv(file="C://Rcodes/insur.csv",header=TRUE)
    ```

- Components of a data frame can be extracted

  by name: `B.dat$age`

  by index: `A.dat[,2]`

  made directly available by `attach(A.dat)`

§3.6. Functions related to the computation of distributions

- There are four sets of functions `d*, p*, q*, r*` for the computation of probability density function, cumulative distribution function, quantiles, and generation of random samples respectively.
- For a particular distribution, the `*` is replace by the abbreviation of the name of that distribution. For example, for Normal distribution:

```
dnorm(x, mean = 0, sd = 1)
pnorm(q, mean = 0, sd = 1, lower.tail = TRUE)
qnorm(p, mean = 0, sd = 1, lower.tail = TRUE)
rnorm(n, mean = 0, sd = 1)
```

  where, by default, `lower.tail = TRUE`, it computes the lower tail probability or quantile. If upper tail probability and quantiles are needed, specify `lower.tail = FALSE`
- Abbreviations for t-, F- and χ^2-distributions are, respectively, `t, f, chisq`. For the `p*` functions, their usages are:

```
pt(q,df), pf(q,df1,df2), pchisq(q,df)
```

§3.7. R functions for regression analysis

The main R function used for regression analysis is `lm`.

```
lm(formula, data, subset, weights, na.action,
   method="qr", model=RUE, x=FALSE, y=FALSE, qr=TRUE,
   singular.ok=TRUE, contrasts=NULL, offset, ...)
```

Functions for extracting fitted results:

```
summary
anova
vcov
```

§3.8. Formula specification

- The `formula` argument of `lm` takes the form `y~formula` where formula specifies the regression function:
  - x1+x2: $\beta_0 + \beta_1 x_1 + \beta_2 x_2$.
  - -1+x1+x2: $\beta_1 x_1 + \beta_2 x_2$.
  - `x1*x2`, or x1+x2+x1:x2: $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \gamma x_1 x_2$.
  - `x1*x2+x1*x3+x2*x3`:

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \gamma_{12} x_1 x_2 + \gamma_{13} x_1 x_3 + \gamma_{23} x_2 x_3.$$

- Special functions.
  - `offset()`: if a variable is specified as `offset(x)`, the coefficient of the variable is fixed as 1.
  - `I()`: if an operation, say, `x1*x2`, is specified as `I(x1*x2)`, it gives a new predictor, the component-wise product of x1 and x2, it is different from the specification `x1*x2`.